* Homogeneous Differential Equations :-

A Differential equation of the form

$$\frac{dy}{dx} = \frac{M(x,y)}{N(x,y)} \quad \text{---(1)}$$

is called a homo. eq^{n} if $M(x,y)$ and $N(x,y)$ are homo. functions of the same degree.

- Eq^{n} (1) can be reduced to variable - separable form by putting $y = vx$.

$$\therefore \frac{dy}{dx} = v + x\frac{dv}{dx}.$$

- Eq^{n} (1) reduces to

$$v + x\frac{dv}{dx} = \frac{M(x,vx)}{N(x,vx)} = g(v).$$

$$x\frac{dv}{dx} = g(v) - v$$

$$\therefore \frac{dv}{g(v)-v} = \frac{dx}{x}.$$

- This eq^{n} is in variable separable form and can be solved by integrating.

$$\therefore \int \frac{dv}{g(v)-v} = \int \frac{dx}{x} + c.$$

- After integrating and replacing v by $\frac{y}{x}$, we get the sol^{n} of eq^{n} (1).

* Note :-

Homogeneous Function : A function $f(x,y,z)$ is

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Said to be homogeneous function of degree n, if for any positive number t, $f(xt, yt, zt) = t^n f(x, y, z)$, where n is a real number.

* Examples:-

① $x(x-y)\,dy + y^2\,dx = 0$.

→

$$\frac{dy}{dx} = \frac{-y^2}{x^2 - xy} = \frac{M(x,y)}{N(x,y)} \quad \text{---(1)}$$

- The eq^n is homogeneous since M and N are of same degree 2.

∴ Let, $y = vx$.

$$\therefore \frac{dy}{dx} = v + x\frac{dv}{dx}.$$

$$\therefore v + x\frac{dv}{dx} = \frac{-v^2x^2}{x^2(1-v)} = \frac{-v^2}{1-v}.$$

$$\therefore x\frac{dv}{dx} = -v - \frac{v^2}{1-v} = -\left(v + \frac{v^2}{1-v}\right)$$

$$= -\left(\frac{v - v^2 + v^2}{1-v}\right)$$

$$= -\frac{v}{1-v}$$

$$\therefore \left(\frac{1-v}{v}\right)dv = -\frac{1}{x}\,dx$$

$$\therefore \left(\frac{1}{v} - 1\right)dv = -\frac{1}{x}\,dx.$$

Integrating both the sides,

$$\log v - v = -\log x + \log c$$

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$\therefore v = \log v + \log x - \log c$

$\therefore \frac{y}{x} = \log \frac{v \cdot x}{c}$

$y = x \cdot \log \frac{y}{c}$.

(2) $x^2 y\,dx - (x^3 + xy^2)\,dy = 0$.

→

$$\frac{dy}{dx} = \frac{x^2 y}{x^3 + xy^2} = \frac{xy}{x^2 + xy} = \frac{M(x,y)}{N(x,y)} \quad \text{—— (1)}$$

- The eq^n is homo. since, M and N are of the same degree 2.

- Let, $y = vx$.

$\therefore \frac{dy}{dx} = v + x\frac{dv}{dx}$.

From (1),

$$v + x\frac{dv}{dx} = \frac{x(vx)}{x^2 + x(vx)} = \frac{x^2 v}{x^2(1+v)} = \frac{v}{1+v}.$$

$$\therefore x\frac{dv}{dx} = \frac{v}{1+v} - v = \frac{v - v(1+v)}{1+v}$$

$$\therefore x\frac{dv}{dx} = \frac{-v^2}{1+v}.$$

$$\therefore \frac{1+v}{v^2}\,dv = -\frac{dx}{x}$$

$$\therefore \left(\frac{1}{v^2} + \frac{1}{v}\right) dv = -\frac{dx}{x}.$$

Integrating both the sides,

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$$-\frac{1}{v} + \log v = -\log x + \log c$$

$$\therefore \log v + \log x = \frac{1}{v} + \log c$$

$$\therefore = \frac{1}{v}\log e + \log c$$

$$\therefore \log vx = \log e^{yv} + \log c.$$

$$\therefore vx = ce^{yv}.$$

$$\therefore y = ce^{x/y}.$$

* Non homogeneous Differential Equations:-

A differential eq^{n} of the form

$$\frac{dy}{dx} = \frac{a_1x + b_1y + c_1}{a_2x + b_2y + c_2} \quad \text{---} \quad (1)$$

is called a non homo. eq^{n} where a_1, b_1, c_1, a_2, b_2 are all constants.

- These eq^{ns} are classified into two parts.

Case-I:- If $\frac{a_1}{a_2} = \frac{b_1}{b_2} = m$

$$\therefore a_1 = ma_2, \quad b_1 = mb_2.$$

then eq^{n} (1) reduces to

$$\frac{dy}{dx} = \frac{ma_2 \cdot x + mb_2y + c_1}{a_2x + b_2y + c_2} = \frac{m(a_2x + b_2y) + c_1}{a_2x + b_2y + c_2}$$

--- (2)

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Putting $a_2x + b_2y = t$,

$$\therefore a_2 + b_2 \frac{dy}{dx} = \frac{dt}{dx}.$$

∴ From (2), ∴ Eq^n (1) reduces to variable-separable form and can be solved using the method of variable separable eq^n.

* Examples:-

① Solve $(x+y-1)\,dx + (2x+2y-3)\,dy = 0$.

→

$$\frac{dy}{dx} = -\frac{x+y-1}{2x+2y-3} = \frac{-x-y+1}{2x+2y-3} \quad \text{——— (1)}$$

The eq^n is non-homo. and $\frac{a_1}{a_2} = \frac{-1}{2}$

$$\frac{b_1}{b_2} = \frac{-1}{2}$$

$$\therefore \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{-1}{2}.$$

Let, $x+y = t$

$$\therefore 1 + \frac{dy}{dx} = \frac{dt}{dx}$$

$$\therefore \frac{dy}{dx} = \frac{dt}{dx} - 1.$$

∴ From (1),

$$\frac{dt}{dx} - 1 = \frac{-t+1}{2t-3}$$

$$\therefore \frac{dt}{dx} = \frac{-t+1}{2t-3} + 1$$

$$= \frac{-t+1+2t-3}{2t-3} = \frac{t-2}{2t-3}$$

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$\therefore \left(\frac{2t-3}{t-2}\right) \left(\frac{2t-3}{t-2}\right) dt = dx.$

$\therefore \left(2 + \frac{1}{t-2}\right) dt = dx.$

Integrating both the sides,

$\int \left(2 + \frac{1}{t-2}\right) dt = \int dx$

$\therefore 2t + \log(t-2) = x + c$

$\therefore 2(x+y) + \log(x+y-2) = x + c.$

$\therefore 2x + 2y - x + \log(x+y-2) = c$

$\therefore x + 2y + \log(x+y-2) = c.$

(2) $(x+y)\,dx + (3x+3y-4)\,dy = 0$, $y(1) = 0$.

$\rightarrow$

$$\frac{dy}{dx} = \frac{-x-y}{3x+3y-4} \quad \text{---(1)}$$

The eq^{n} is non-homogeneous and

$\frac{a_1}{b_1} = \frac{a_2}{b_2} \Rightarrow \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{-1}{3}.$

Let, $x + y = t$

$1 + \frac{dy}{dx} = \frac{dt}{dx}.$

$\therefore \frac{dy}{dx} = \frac{dt}{dx} - 1.$

$\therefore$ From (1),

$$\frac{dt}{dx} - 1 = \frac{-t}{3t-4}$$

$$\therefore \frac{dt}{dx} = 1 - \frac{t}{3t-4} = \frac{2t-4}{3t-4}$$

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$\therefore \left(\frac{3t-4}{2t-4}\right) dt = dx.$

$\therefore \frac{1}{2}\left(\frac{3+2}{t-2}\right) dt = dx$

Integrating both the sides,

$\frac{1}{2}\int \left(\frac{3+2}{t-2}\right) dt = \int dx$

$\therefore \frac{1}{2}[3t + 2\log|(t-2)|] = x + c$

$\therefore 3(x+y) + 2\log|(x+y-2)| = 2x + 2c$

$\therefore x + 3y + 2\log|(x+y-2)| = k$, where $k = 2c$.

- Given, $y(1) = 0$

Putting $x = 1, y = 0$ in above eq^n.

$\therefore 1 + 2\log|-1| = k$

$\therefore 1 + 2\log 1 = k$

$\therefore k = 1.$

Hence, the sol^n is $x + 3y + 2\log|(x+y-2)| = 1.$

* Case - II :- If $\frac{a_1}{a_2} = \frac{b_1}{b_2}$.

Then substituting $x = X+h$, $y = Y+k$ in eq^n (1).

$\therefore \frac{dy}{dx} = \frac{dY}{dX}.$

From (1), $\frac{dY}{dX} = \frac{a_1(X+h) + b_1(Y+k) + c_1}{a_2(X+h) + b_2(Y+k) + c_2}$

$= \frac{(a_1X + b_1Y) + (a_1h + b_1k + c_1)}{(a_2X + b_2Y) + (a_2h + b_2k + c_2)}$ — (3)

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Choosing h, k such that,
$a_1h + b_1k + c_1 = 0$, $a_2h + b_2k + c_2 = 0$.

$\therefore$ Eq^n (3) reduces to

$$\frac{dy}{dx} = \frac{dY}{dX} = \frac{a_1X + b_1Y}{a_2X + b_2Y}.$$

Which is a homo. eq^n and can be solved using the method of homo. eq^n.

- Finally, substituting, $X = x - h$, $Y = y - k$, we get sol^n of eq^n (1).

* Examples: -

(1) Solve $(x + 2y)\,dx + (y - 1)\,dy = 0$.

→

$$\frac{dy}{dx} = -\frac{x + 2y}{y - 1} \quad \text{——(1)}$$

- The eq^n is non homo. and $\frac{-1}{0} \neq \frac{-2}{1}$

Let, $x = X + h$, $y = Y + k$

$\therefore dx = dX$, $dy = dY$.

$$\therefore \frac{dy}{dx} = \frac{dY}{dX}.$$

-Substituting in (1),

$$\frac{dY}{dX} = \frac{-(X+h) - 2(Y+k)}{(Y+k) - 1}$$

$$= \frac{(-X - 2Y) + (-h - 2k)}{Y + (k-1)} \quad \text{——(2)}$$

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Choosing h, k such that $-h-2k=0$, $k-1=0$ — (3)

$\therefore k=1$, $h=-2$.

Substituting (3) in (2),

$$\therefore \frac{dY}{dX} = \frac{-X-2Y}{Y} \quad \text{— (4)}$$

Which is a homo. eqⁿ

Let, $Y = vX$.

$$\therefore \frac{dY}{dX} = v + X \cdot \frac{dv}{dX}.$$

From (4),

$$v + X\frac{dv}{dX} = \frac{-X-2vX}{vX}$$

$$= \frac{-1-2v}{v}$$

$$\therefore X\frac{dv}{dX} = \frac{-1-2v}{v} - v$$

$$= \frac{-1-2v-v^2}{v}$$

$$= -\frac{(v+1)^2}{v}$$

$$\therefore \frac{v}{(v+1)^2}\,dv = -\frac{dx}{X}.$$

$$\left[\frac{1}{v+1} - \frac{1}{(v+1)^2}\right] dv = -\frac{dx}{X}.$$

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Integrating both the sides,

$$\int \frac{1}{v+1}\,dv - \int \frac{1}{(v+1)^2}\,dv = -\int \frac{dx}{x}.$$

$$\therefore \log(v+1) + \frac{1}{v+1} = -\log x + c$$

$$\therefore \log\left(\frac{y}{x}+1\right) + \frac{1}{\frac{y}{x}+1} = -\log x + c$$

$$\therefore \log\left(\frac{y+x}{x}\right) + \frac{x}{y+x} = -\log x + c$$

$$\therefore \log(y+x) - \log x + \frac{x}{y+x} = -\log x + c$$

$$\therefore \log(y+x) + \frac{x}{y+x} = c.$$

Now, $X = x-h = x+2$

$Y = y-k = y-1$

Hence, the G.S. is

$$\log(y-1+x+2) + \frac{x+2}{y-1+x+2} = c$$

$$\therefore \log(x+y+1) + \left(\frac{x+2}{x+y+1}\right) = c.$$

(2) $\frac{dy}{dx} = \frac{2x-5y+3}{2x+4y-6}$.

→

The eqⁿ is non-homo. and $\frac{2}{2} \neq \frac{-5}{4}$

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Let, $x = X+h$, $y = Y+k$.

$\therefore dx = dX$, $dy = dY$.

$\therefore \frac{dy}{dx} = \frac{dY}{dX}$.

$\therefore$ The given eq^n becomes

$$\frac{dY}{dX} = \frac{2(X+h) - 5(Y+k) + 3}{2(X+h) + 4(Y+k) - 6}$$

$$= \frac{(2X-5Y) + (2h-5k+3)}{(2X+4Y) + (2h+4k-6)} \quad \text{——(1)}$$

Choosing h, k such that $2h-5k+3=0$, $2h+4k-6=0$ ——(2)

$\therefore h = k = 1$.

Substituting (2) in eq^n (1),

$$\therefore \frac{dY}{dX} = \frac{2X-5Y}{2X+4Y} \quad \text{——(3)}$$

Which is a homo. eq^n.

Let, $Y = vX$.

$$\therefore \frac{dY}{dX} = v + X\cdot\frac{dv}{dX}.$$

$\therefore$ From (3),

$$v + X\frac{dv}{dX} = \frac{2X-5vX}{2X+4vX} = \frac{2-5v}{2+4v}$$

$$\therefore X\frac{dv}{dX} = \frac{2-5v}{2+4v} - v$$

$$= \frac{2-5v-2v-4v^2}{2+4v}.$$

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$$\therefore x\frac{dv}{dx} = \frac{-4v^2-7v+2}{2+4v}$$

$$\therefore \frac{2+4v}{4v^2+7v-2}\,dv = -\frac{dx}{x} \qquad \therefore x\frac{dv}{dx} =$$

$$\therefore \frac{2+4v}{4v^2+7v-2}\,dv = -\frac{1}{x}\,dx. \qquad \text{---(4)}$$

Now, $\frac{2+4v}{4v^2+7v-2} = \frac{A}{(4v-1)} + \frac{B}{v+2}$

$\therefore 2+4v = A(v+2) + B(4v-1)$

Take $v = -2$

$\therefore 2+4(-2) = 0 + B(-9)$

$\therefore -6 = -9B$

$\therefore B = \frac{2}{3}$

Take $v = \frac{1}{4}$,

$\therefore 2+4\left(\frac{1}{4}\right) = A\left(\frac{1}{4}+2\right) + B(0)$

$\therefore 3 = A\left(\frac{9}{4}\right)$

$\therefore A = \frac{4}{3}$

$$\therefore \frac{2+4v}{(4v-1)(3v+2)} = \frac{4}{3(4v-1)} + \frac{2}{3(v+2)}$$

Substituting in eqⁿ (4),

$$\left[\frac{4}{3(4v-1)} + \frac{2}{3(v+2)}\right] dv = -\frac{dx}{x}.$$

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Integrating both the sides,

$$\int \left[\frac{4}{3(4v-1)} + \frac{2}{3(v+2)} \right] dv = \frac{-dx}{x}.$$

$$\therefore \frac{4}{3} \log \frac{(4v-1)}{4} + \frac{2}{3} \log (v+2) = -\log x + \log c$$

$$\therefore \frac{1}{3} \log (4v-1)(v+2)^2 = \log \frac{c}{x}$$

$$\therefore \log (4v-1)^{1/3} (v+2)^{2/3} = \log \frac{c}{x}$$

$$\therefore (4v-1)^{1/3} (v+2)^{2/3} = \frac{c}{x}.$$

$$\therefore \left(\frac{4y}{x} - 1 \right)^{1/3} \left(\frac{y}{x} + 2 \right)^{2/3} = \frac{c}{x}.$$

$$\therefore (4y-x)^{1/3} (y+2x)^{2/3} = c$$

$$\therefore (4y-x)(y+2x)^2 = c^3 = k$$

$X = x - h = x - 1$

$Y = y - k = y - 1$

Hence, the G.S. is

$$(4y - x - 3)(y + 2x - 3)^2 = k.$$

Non-Exact Eq^{n} :-

Case I :-

If the D.E. is of the type

$$x^{m_1} y^{n_1} (a_1 y\, dx + b_1 x\, dy) + x^{m_2} y^{n_2} (a_2 y\, dx + b_2 x\, dy) = 0.$$

Then I.F. $= x^h y^k$, where

$$\frac{m_1 + h + 1}{a_1} = \frac{n_1 + k + 1}{b_1} \quad \& \quad \frac{m_2 + h + 1}{a_2} = \frac{n_2 + k + 1}{b_2}.$$

Solving these two eq^{n}, we get the values of h & k.

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* Examples:-

① Solve $x(4y\,dx + 2x\,dy) + y^3(3y\,dx + 5x\,dy) = 0$

→

$x y^0 (4y\,dx + 2x\,dy) + x^0 y^3 (3y\,dx + 5x\,dy) = 0$

$m_1 = 1, n_1 = 0, m_2 = 0, n_2 = 3$

$a_1 = 4, b_1 = 2,$

$a_2 = 3, b_2 = 5$

Now, $\frac{m_1 + h + 1}{a_1} = \frac{n_1 + k + 1}{b_1}$

$\therefore \frac{1 + h + 1}{4} = \frac{0 + k + 1}{2}$

$\therefore 2(2 + h) = 4(k + 1)$

$\therefore 6 + 3h = 4k + 4$

$\therefore h + 2 = 2(k + 1)$

$h + 2 = 2k + 2$

$\therefore h = 2k$ — (1)

and $\frac{m_2 + h + 1}{a_2} = \frac{n_2 + k + 1}{b_2}$

$\therefore \frac{0 + h + 1}{3} = \frac{3 + k + 1}{5}$

$\therefore 5(h + 1) = 3(k + 4)$

$\therefore 5h + 5 = 3k + 12$

$\therefore 5h - 3k = 7$

From (1), $5(2k) - 3k = 7$

$10k - 3k = 7$

$k = 1, \quad h = 2$

$\therefore$ I.F. $= x^h y^k = x^2 y$

Multiplying the D.E. by the I.F.,

$x^2 y \cdot x(4y\,dx + 2x\,dy) + x^2 y \cdot y^3 (3y\,dx + 5x\,dy) = 0$

$\therefore (4x^3 y^2\,dx + 2x^4 y\,dy) + (3x^2 y^5\,dx + 5x^3 y^4\,dy) = 0$

$\therefore (4x^3 y^2 + 3x^2 y^5)\,dx + (2x^4 y + 5x^3 y^4)\,dy = 0$

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$\therefore M = 4x^3y^2 + 3x^2y^5$, $N = 2x^4y + 5x^3y^4$

$\therefore \frac{\partial M}{\partial y} = 8x^3y + 15x^2y^4$, $\frac{\partial N}{\partial x} = 8x^3y + 15x^2y^4$

$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \Rightarrow$ Eq^n is exact.

$\therefore$ The G.S. is

$\int M\,dx + \int$ (Terms of N not containing x) $dy = c$

y-const

$\therefore \int (4x^3y^2 + 3x^2y^5)\,dx + \int 0\,dy = 0.$

$\therefore x^4y^2 + x^3y^5 = c.$

(2) Solve: $(x^7y^2 + 3y)\,dx + (3x^8y - x)\,dy = 0.$

$\Rightarrow$

$M = x^7y^2 + 3y$, $N = 3x^8y - x.$

$\frac{\partial M}{\partial y} = 2x^7y + 3$, $\frac{\partial N}{\partial x} = 24x^7y - 1$

$\therefore \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x} \Rightarrow$ Not exact.

- Rewriting the eq^n,

$(x^7y^2\,dx + 3x^8y\,dy) + (3y\,dx - x\,dy) = 0$

$\therefore x^7y\,(y\,dx + 3x\,dy) + x^0y^0\,(3y\,dx - x\,dy) = 0$

$\therefore m_1 = 7,\ n_1 = 1,\ a_1 = 1,\ b_1 = 3,\ m_2 = 0,\ n_2 = 0,$
$a_2 = 3,\ b_2 = -1$

$\therefore \frac{m_1 + h + 1}{a_1} = \frac{m_2 + k + 1}{b_1}$

$\therefore \frac{7 + h + 1}{1} = \frac{1 + k + 1}{3}$

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$\therefore 3(8+h) = k+2$

$\therefore 24 + 3h = k+2$

$\therefore 3h - k = -22$ —— (1)

$$\frac{m_2 + h + 1}{a_2} = \frac{n_2 + k + 1}{b_2}$$

$$\therefore \frac{0+h+1}{3} = \frac{0+k+1}{-1}$$

$\therefore -h-1 = 3k+3$

$\therefore h + 3k = -4$ —— (2)

From (1) & (2), $h = -7$, $k = 1$

$\therefore$ I.F. $= x^{-7}y$

Multiplying the D.E. by I.F.,

$x^{-7}y\,(x^7y^2 + 3y)\,dx + x^{-7}y\,(3x^8y - x)\,dy = 0$

$(y^3 + 3yx^{-7})\,dx + (3xy^2 - x^{-6})\,dy = 0$

$\therefore M_1 = y^3 + 3yx^{-7}$, $N_1 = 3xy^2 - x^{-6}$

$\frac{\partial M_1}{\partial y} = \frac{\partial N_1}{\partial x} \Rightarrow$ eq^n is exact.

Sol^n is $xy^3 - \frac{x^{-6}y^2}{2} = c$

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